Digital Twin Anchored Generation: Adaptive Querying and Phenotypic Refinement

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***Abstract*—Digital Twin Anchored Generation (DTAG) adaptively refines a *predictive phenotype*: conditional distributions over a respondent's unasked survey items. A Large Science Model (LSM) predicts responses, and an optimizer selects questions by their expected displacement of a pole-relative geometric index. In native GSS 2024 experiments, six pinned personas generated 960 adaptive 20-question trajectories. All 27 reference-pole items were excluded from selection. Right- and left-seeking policies produced directional index shifts; 2,880 fixed-set replays retained these directions under reversed and shuffled orders, without a detected systematic mean order effect. We describe a prospective randomized human study testing whether the predicted shifts transfer to actual responses and whether questioning also affects subsequent reports.**

***Index Terms*—Predictive phenotypes, large science models, conditional response modeling, digital twins, survey methodology, opinion dynamics, adaptive questioning, generative simulation.**

I. Introduction

Survey answers are statistically coupled: each response changes predictions for questions not yet asked. We investigate whether questions can be selected to refine a respondent's inferred opinion profile in a prescribed direction.

Digital Twin Anchored Generation (DTAG) represents observed answers as a partial state and uses a Large Science Model (LSM) to predict unasked responses. This distributional description is the *predictive phenotype*. A distance to two reference opinion configurations defines a pole-relative index; DTAG selects successive questions according to their expected effect on this index.

We derive the policy and evaluate it using six pinned GSS 2024 personas. Adaptive simulations test directional refinement; fixed-set replays examine sensitivity to order and continued optimization. All pole-defining items are withheld from questioning. We also outline a prospective human experiment to test directional prediction from real answers and possible effects on subsequent judgments.

II. Related Work

Attitude dependencies need not align on one ideological axis [1], [2]. Partisan sorting and social influence shape these dependencies [3], [4]; cognitive consistency, wording, and order can affect reported attitudes [5]–[7]. This motivates separating information revealed by a question from any change induced by asking it.

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Opinion-dynamics models often use low-dimensional transitions [8]–[10], while experiments on exposure and contact report heterogeneous effects [11]–[13]. Generative agents and prompted survey personas can produce plausible responses [14], [15]. DTAG instead combines learned high-dimensional survey conditionals with an explicit question-selection objective.

III. Large Science Models and the Predictive Phenotype

A. Conditional dependency model

Let $X = (X_1, \ldots, X_n)$ be categorical survey responses, with alphabets $\mathcal{A}_i$ and survey context c. An LSM comprises conditionals

$$\Phi_c = \{\phi_{i,c}(a \mid x_{-i}) : i = 1, \ldots, n\}, \tag{1}$$

estimated by conditional inference trees [16]–[18]. Their dependencies are predictive, not causal; the conditionals need not form a coherent joint law [19].

A respondent state $s = \{X_j = a_j : j \in S\}$ holds observed responses. Unanchored variables remain missing in $\widetilde{x}_{-v}(s)$; each native tree averages over its missing branches. For unasked $v \notin S$,

$$p_\Phi(a \mid s, v) = \phi_{v,c}(a \mid \widetilde{x}_{-v}(s)). \tag{2}$$

The vector

$$\mathcal{B}_\Phi(s) = \big(p_\Phi(\cdot \mid s, v)\big)_{v \notin S} \tag{3}$$

is the *predictive phenotype*: conditional distributions and uncertainty for unasked responses.

B. Geometry as predictive equivalence

Conditional predictions induce a distance between complete or partial states x, y:

$$\theta_\Phi(x, y) = \frac{1}{n} \sum_{i=1}^{n} \sqrt{D_{\mathrm{JS}}(\phi_{i,c}(\cdot \mid x_{-i}), \phi_{i,c}(\cdot \mid y_{-i}))}. \tag{4}$$

Here D_{JS} denotes Jensen–Shannon divergence. The resulting predictive pseudometric distinguishes states by their conditional consequences.

Prior LSM reconstruction reduced mean distance distortion by 54.9% (GSS) and 60.9% (Eurobarometer) with 80% of responses masked [18].

For GSS models with registered left and right reference states s_L, s_R, DTAG reports

$$I_\Phi(s) = \frac{\theta_\Phi(s_L, s) - \theta_\Phi(s_R, s)}{\theta_\Phi(s_L, s_R)}. \tag{5}$$

The reference poles specify opposing responses on 27 GSS 2024 items covering policy, civil liberties, spending, and religion (Table I) [18]. Other coordinates are unspecified; I_Φ measures predictive position rather than self-reported ideology.

TABLE I

GSS 2024 reference poles. TWENTY-SEVEN PAIRED RESPONSES DEFINE s_L, s_R, WITH $I_\Phi = -1, +1$; ALL OTHER COORDINATES ARE UNSPECIFIED.

Item	Survey topic	Left	Right
abany	Abortion on request	yes	no
abdefect	Abortion: fetal defect	yes	no
abnomore	Abortion: no more children	yes	no
abpoor	Abortion: low income	yes	no
abrape	Abortion after rape	yes	no
absingle	Abortion: unmarried	yes	no
bible	Bible interpretation	ancient book	inspired word
colcom	Communist as college teacher	not fired	yes, fired
confed	Trust in federal government	a great deal	hardly any
conlabor	Trust in organized labor	a great deal	hardly any
grass	Marijuana legalization	should be legal	should not be legal
gunlaw	Permit to purchase gun	favor	oppose
libcom	Communist books in library	not remove	remove
libmslm	Muslim cleric books in library	not remove	remove

Item	Survey topic	Left	Right
natarms	Military spending	too much	about right
natenvir	Environmental spending	too little	about right
natfare	Welfare spending	too little	too much
natsoc	Social Security spending	too little	about right
owngun	Guns in household	no	yes
pillok	Contraception access	strongly agree	disagree
polabuse	Police strike insulting citizen	yes	no
pray	Frequency of prayer	never	several times a day
prayer	Bible prayer in public schools	approve	disapprove
religen	Personal religiosity	no religion	strong
rowngun	Respondent owns a gun	no	yes
shotgun	Owns a shotgun	no	yes
spkcom	Communist public speech	yes, allowed to speak	not allowed

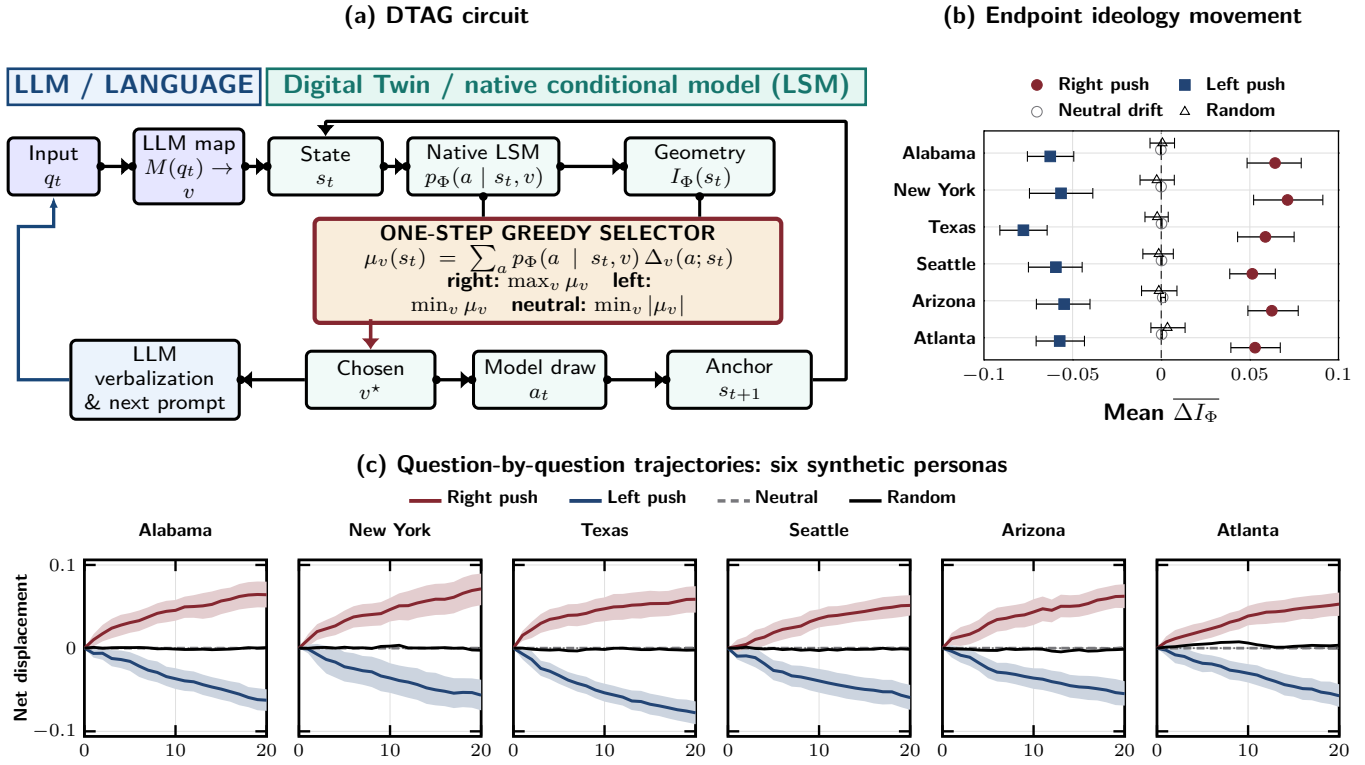


Fig. 1. **DTAG architecture and native adaptive results.** (a) LSM-anchored responses and directional question selection; language mapping is a separate interface. (b) Mean endpoint displacement across six personas with 95% bootstrap intervals. (c) Mean 20-question trajectories over 40 repetitions per persona and policy, with pointwise 95% intervals (right: red; left: blue; neutral: dashed gray; random: black).

IV. Adaptive Question Selection

A semantic interface can map a natural-language question q to a survey item $v \in M(q)$ and verbalize its answer. The native experiments use item identifiers directly without the LLM layer. For a selected question, DTAG samples a response and updates the predictive phenotype:

$$A_t \sim p_\Phi(\cdot \mid s_t, v_t), \tag{6}$$

$$s_{t+1} = s_t \cup \{v_t = A_t\}, \tag{7}$$

$$\mathcal{B}_{t+1} = \mathcal{B}_\Phi(s_{t+1}). \tag{8}$$

The initial persona anchors remain fixed. The optimizer chooses the question, not its answer (Fig. 1).

For candidate item v and answer a, define

$$\Delta_v(a; s) = I_\Phi(s \cup \{v = a\}) - I_\Phi(s), \tag{9}$$

$$\mu_v(s) = \sum_{a \in \mathcal{A}_v} p_\Phi(a \mid s, v)\Delta_v(a; s), \tag{10}$$

$$\sigma_v^2(s) = \sum_a p_\Phi(a \mid s, v)(\Delta_v(a; s) - \mu_v(s))^2. \tag{11}$$

Right- and left-seeking policies maximize and minimize $\mu_v(s)$; neutral minimizes $|\mu_v(s)|$, and random samples an eligible item. By construction,

$$\mathbb{E}[I_\Phi(s_{t+1}) - I_\Phi(s_t) \mid s_t, v_t] = \mu_{v_t}(s_t).$$

The accumulated displacement depends on successive selected items and sampled answers.

The effect on an unasked item j after observing (v, a) is

$$R_j(s; v, a) = D_{\mathrm{JS}}\big(p_\Phi(\cdot \mid s, j), p_\Phi(\cdot \mid s \cup \{v = a\}, j)\big). \quad (12)$$

A weighted sum $\mathcal{P}_\Phi(s; v, a) = \sum_{j \notin S \cup \{v\}} w_j R_j(s; v, a)$ measures redistribution across the remaining items. Unlike text retrieval [20], DTAG maintains an explicit conditional response state. The public implementation includes a web application (https://github.com/zeroknowledgediscovery/DTAG/tree/main/webapp).

V. Refinement and Possible Response Effects

Under *phenotypic refinement*, successive answers restrict predictions for unasked items. For held-out responses Y, the expected information from asking v is

$$\mathcal{G}(v \mid s) = H(Y \mid s) - \mathbb{E}_{A_v|s}[H(Y \mid s, v = A_v)]. \quad (13)$$

Under a coherent joint model this is conditional mutual information, distinct from directional displacement $\mu_v(s)$. Held-out response prediction evaluates refinement directly.

A. Possible response effects.

Questioning a person may change later reports. A response–transition kernel separates an observed answer from a possible latent-state update:

$$K_q(a, z' \mid z) = O_q(a \mid z)\, T_q(z' \mid z, a). \quad (14)$$

Passive observation has $T_q(z' \mid z, a) = \delta_z(z')$; shifts in salience or belief can induce nontrivial transitions. A common held-out battery and delayed measurement can investigate these effects.

B. Order dependence.

For fixed realized answers, the terminal index is order-independent; sequentially generated answers need not be. A coherent joint distribution satisfies

$$P(a \mid s, v)P(b \mid s, v = a, w) = P(b \mid s, w)P(a \mid s, w = b, v). \quad (15)$$

Separately fitted LSM conditionals may violate this identity [19]. Fixed-set replays evaluate its consequences for native response sequences.

VI. Native GSS 2024 Computational Experiments

A. Adaptive questioning from pinned persona states

We fixed six initial GSS 2024 persona states (Table II) and generated 40 independent response sequences per persona and policy (right, left, neutral, random). Each sequence comprised 20 questions with responses sampled from the native LSM; no LLM mapping or verbalization was used. *All 27 pole-defining items were excluded from selection in every policy* (Table I). Thus 960 adaptive trajectories contain 19,200 non-pole responses; repetitions share initial anchors but have independent sampled answers.

Across all six personas, right- and left-seeking policies moved the index in their intended directions (Fig. 1). As pole-defining questions were excluded, the movement results from conditioning on other survey items, not directly asking those used to define the reference index. Neutral and random policies provide comparisons.

B. Question selection as observed paths

Question identities are summarized in Table III. The 18 displayed heatmaps contain 161 distinct items across right, left, and neutral selection (Fig. 2). A shared categorical ordering approximately minimizes

$$\mathcal{C}(\pi) = \sum_{p,u,t} \sum_{r<r'} \big|\pi(v_{purt}) - \pi(v_{pur't})\big|, \quad (16)$$

Here u indexes policies. Heatmap colors identify items, not ideological position; 67 further items occurred only in random selection.

C. Fixed-set replay: robustness to question ordering

For each persona and right/left/neutral policy, we froze the 20-item sequence from adaptive replicate zero, giving 18 fixed sets. Each set was replayed in original, reversed, and two shuffled orders with 40 fresh sampled responses per order (2,880 runs). Matched question-specific random variates reduced contrast variance.

The direction persisted across replays (Fig. 2): right-derived sets yielded mean displacement $+0.0381$ (original) and $+0.0377$ (reversed); left-derived sets yielded -0.0399 and -0.0410. In both orders, all six persona-specific 95% bootstrap intervals excluded zero in the prescribed direction. Shuffles followed the same pattern; neutral sets remained near zero.

All 54 pointwise bootstrap intervals for paired mean between-order contrasts included zero; no systematic mean order effect was detected. Selected question sets therefore retained directional displacement without continuous reoptimization.

In historical ten-question runs, baseline indices differed across environments despite matched persona, map, and pole hashes; native binaries were not compared. Their absolute indices are not combined with the present experiment.

VII. Prospective Human Validation

The native experiments establish **H1**: directional question selection yields reproducible shifts of the inferred phenotype under the learned conditional model. The prospective human test addresses **H2**: whether the policy predicts the direction of index movement when questions are answered by people.

Prolific respondents will complete a common baseline battery and be randomized to right, left, neutral, or random question selection. All arms receive matched-length questioning. The primary outcome is $I_\Phi(s_T) - I_\Phi(s_0)$ from actual responses; prespecified arm contrasts test the directional prediction. Question-level probability scores assess calibration. A common held-out post-battery tests whether questioning history also affects later reports, beyond the information revealed; optional recontact [21] assesses persistence. A pilot will set sample size, with consent and debriefing.

TABLE II
PINNED GSS 2024 SYNTHETIC PERSONAS AND BASELINE POLE-RELATIVE INDEX.

Persona	**Pinned description**	**Initial $I_\Phi(s_0)$**
Alabama	45-year-old rural Alabama white man; married, two children; full-time farmer and veteran; high school; conservative, strong Republican; Protestant, weekly attendance; daily newspaper.	0.075706
New York	22-year-old New York City white woman; unmarried, no children; full-time retail worker, nonveteran; bachelor's degree; extremely liberal, strong Democrat; nonreligious; daily newspaper.	−0.163807
Texas	61-year-old small-town Texas white woman; married, three children; homemaker, high school; extremely conservative, strong Republican; Protestant, attends services several times weekly; daily newspaper.	0.045379
Seattle	34-year-old Seattle white man; unmarried, no children; employed full time, graduate degree; liberal, weak Democrat; nonreligious; newspaper several times weekly.	−0.119639
Arizona	39-year-old Mexican American man in an Arizona city; married, three children; employed full time, high school; conservative, weak Republican; Catholic, nearly weekly attendance.	−0.020437
Atlanta	47-year-old Atlanta Black woman; divorced, two children; employed full time, bachelor's degree; liberal, strong Democrat; Protestant, weekly attendance; daily newspaper.	−0.030807

Positive values lie nearer the right reference pole; negative values nearer the left.

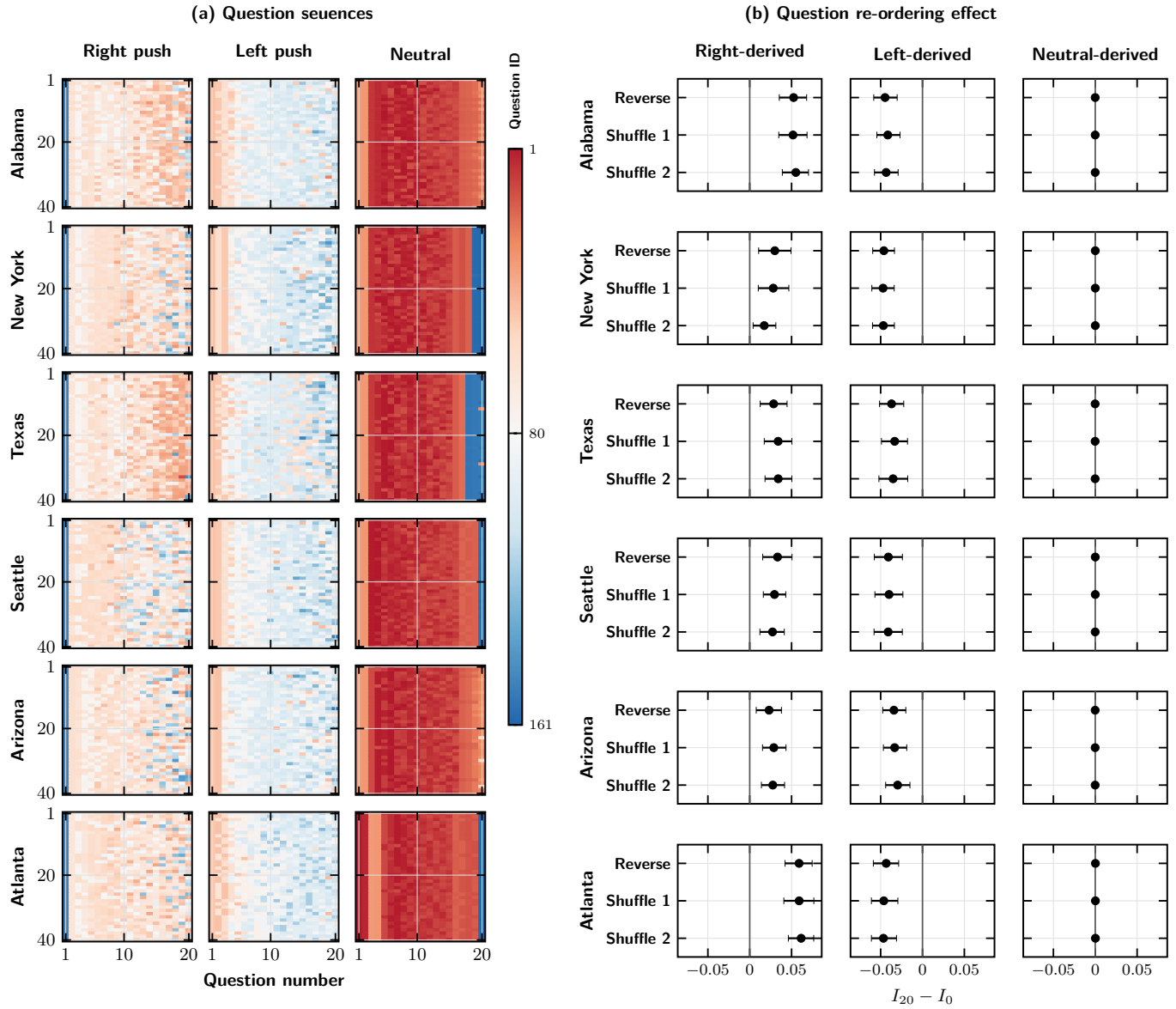


Fig. 2. **Selected questions and fixed-set replays.** Left: question-ID heatmaps for six personas and three selection policies (40 runs each); color denotes item identity. Right: mean index displacement and 95% bootstrap intervals for 20-item sets replayed in original, reversed, and two shuffled orders. Right- and left-derived sets retain their direction; neutral sets remain near zero.

TABLE III
QUESTION SELECTION IN THE NATIVE GSS 2024 EXPERIMENT.

Selection policy	Distinct	Selections	Pole overlap
Right-seeking	100	4,800	0
Left-seeking	75	4,800	0
Neutral	43	4,800	0
Three-policy union	161	14,400	0
All four policies	228	19,200	0

The random policy contributed 4,800 selections; 67 of the 228 distinct items occurred only under random selection.

VIII. Discussion and Conclusion

DTAG uses high-dimensional survey dependencies to select questions that systematically refine a predictive phenotype. The native experiments show directional movement using exclusively non-pole items; fixed-set replays indicate that selected question identities, rather than their precise order, account for much of the directionality. The index depends on the reference poles, candidate pool, and conditional model and should be interpreted as predictive position rather than a self-reported ideology score.

In people, asking questions may also alter the availability or content of later judgments, potentially including more persistent attitude change. The proposed human study tests conditional prediction directly and evaluates such response effects separately.

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